Team 155

Problem B

Two Points to the Goal: A Noise-Conditioned Mixture Density Network for Probabilistic Penalty Kick Prediction

Summary

From two early positions of a penalty kick, we estimate where the ball crosses the goal plane and quantify uncertainty. We integrate gravity, quadratic drag and Magnus lift, then train a noise-conditioned residual mixture density network on airborne trajectories drawn from a mixture of uniform launches and priors informed by tracked flights, including Asai's curve kicks.

For the stated kick at an assumed 5 mm coordinate noise, the model gives a mode of $(Y, Z) = (2.67, 1.79)$ m and joint 50% and 95% region areas of 0.106 and 0.619 m^2. Conditional on airborne crossing, the estimated probability of crossing within a goal rectangle inset by one ball radius is 97.7%.

Of 1,225,000 simulated launches, 1,091,052 cross in flight. On 109,108 held-out crossings the median error is 0.266 m at 5 mm, and joint coverage is 50.1% and 95.1%. Aiming increases simulated high-speed in-frame coverage, with little change in error on shared uniform-prior test flights or on three Asai kicks withheld from the prior. In a separate paired experiment, including rebound and rolling crossings in training reduces median error on post-contact crossings from 0.242 to 0.163 m at 5 mm.

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1 Introduction

1.1 Problem Restatement

Take the penalty spot as the origin and the goal as the rectangle $|Y| < 3.66\,\mathrm{m}$, $0 < Z < 2.44\,\mathrm{m}$ in the plane $x = 11\,\mathrm{m}$ [1]. From $\mathbf{r}(0) = (0, 0, 0.11)\,\mathrm{m}$ and two later observations, the contest asks for the most likely crossing point and its uncertainty. The stated case gives [2]

$$\mathbf{r}(0.05) = (1.50, 0.45, 0.44)\,\mathrm{m}, \qquad \mathbf{r}(0.10) = (2.96, 0.86, 0.74)\,\mathrm{m}. \tag{1}$$

Unknown spin and aerodynamic forces make extrapolation from two points ambiguous. With a fixed start, 5 mm independent coordinate errors give a second-difference acceleration uncertainty of $4.47\,\mathrm{m\,s^{-2}}$. For the stated kick, the same noise gives a median radial deviation of about 0.34 m under quadratic extrapolation (Appendix A).

1.2 Background and Approach

Wind-tunnel tests of non-spinning balls found drag coefficients near 0.43 and 0.15 on either side of the drag crisis, at Reynolds numbers around $2.2\text{–}3.0 \times 10^5$ [3]. A later ball comparison reached 3.3×10^5 [4]. Free-kick fits give $C_D = 0.25\text{–}0.30$. In the same study ten free kicks had speeds of $23.0\text{–}28.3\,\mathrm{m\,s^{-1}}$ at 0.1 s [8]. Trajectory studies find lift coefficients of order 0.2–0.3 under some conditions and estimate aerodynamic curves from tracked flights [6, 7]. These results inform our choices of prior bounds and lift law.

We compare a quadratic extrapolation, a Bayesian reference and an MDN [19] under shared flight dynamics. We fit simulation priors using tracked Asai, ISSIA-3D and DFL trajectories and digitised Barber–Carré data [5, 15, 16, 17, 18]. Held-out simulations assess coverage. A separate source holdout tests transfer to the three Asai kicks.

2 Materials and Methods

2.1 Assumptions and Approximations

Forecasts use the known start, fixed observation times and specified coordinate noise σ. The main endpoint distribution conditions on reaching the goal plane before ground contact, denoted by E. Section 3.7 also considers crossings after ground contact. We report the density mode for the stated case and use the mean for point-error scoring. We score geometric in-frame arrival when the ball centre crosses within $|Y| \le 3.55\,\mathrm{m}$ and $0.11 \le Z \le 2.33\,\mathrm{m}$, allowing one ball radius at each boundary.

A1. Ball. A rigid sphere has mass 0.430 kg, radius 0.110 m and cross-section $A = \pi R^2$ [1]. Buoyancy is omitted (Section 2.2).

A2. Air. The main model uses still air at $\rho = 1.20\,\mathrm{kg\,m^{-3}}$.

A3. Drag. Each flight has a constant C_D in $[0.15, 0.30]$, a range that includes reported free-kick and supercritical wind-tunnel estimates [8, 3].

A4. Lift. Transverse spin determines lift. The slope and cap in Eq. (5) are chosen to match the measured order of magnitude [6, 3].

A5. Spin. We hold spin fixed, following the short-flight approximation discussed by Team 358 [9]. Section 3.6 tests a 0.5 s decay time.

A6. Measurement. The start is exact. The two observations have independent Gaussian coordinate errors with common standard deviation $\sigma \in \{2, 5, 10, 15\}$ mm in evaluation.

A7. Prior. Table 1 gives the common simulation bounds. The Bayesian reference uses independent uniforms and the main training set uses the mixture in Section 2.5.

Table 1: Launch-parameter bounds for the simulation. Uniform reference draws are independent within these bounds.

Parameter	Lower	Upper
v_x $(\mathrm{m\,s^{-1}})$	18	35
v_y $(\mathrm{m\,s^{-1}})$	-12	12
v_z $(\mathrm{m\,s^{-1}})$	0.5	12
$\omega_x, \omega_y, \omega_z$ $(\mathrm{rad\,s^{-1}})$	-60	60
C_D	0.15	0.30

2.2 Physical Model

Let $\mathbf{u} = \mathbf{v} - \mathbf{w}$ be the velocity relative to air, $\mathbf{g} = -g\hat{\mathbf{z}}$ with $g = 9.81\,\mathrm{m\,s^{-2}}$, and $V = 4\pi R^3/3$. The force balance combines weight, quadratic drag, transverse lift and Archimedean buoyancy [10, 11, 12, 13].

$$m\dot{\mathbf{v}} = m\mathbf{g} - \tfrac{1}{2}\rho A C_D |\mathbf{u}|\mathbf{u} + \tfrac{1}{2}\rho A C_L |\mathbf{u}|^2 \hat{\mathbf{n}} - \rho V \mathbf{g}. \tag{2}$$

Buoyancy is 0.066 N upward, or 1.6% of the ball's weight. Omitting it gives

$$\dot{\mathbf{r}} = \mathbf{v}, \tag{3}$$

$$\dot{\mathbf{v}} = \mathbf{g} - \frac{\rho A C_D}{2m}|\mathbf{u}|\mathbf{u} + \frac{\rho A C_L}{2m}|\mathbf{u}|^2\hat{\mathbf{n}}. \tag{4}$$

We use a three-dimensional spin vector $\boldsymbol{\omega}$ and define

$$\hat{\mathbf{n}} = \begin{cases} \dfrac{\boldsymbol{\omega}\times\mathbf{u}}{|\boldsymbol{\omega}\times\mathbf{u}|}, & |\boldsymbol{\omega}\times\mathbf{u}| > 0, \\ \mathbf{0}, & \text{otherwise}, \end{cases} \qquad C_L = \min\!\left(1.4\,\frac{R\,|\boldsymbol{\omega}\times\mathbf{u}|}{|\mathbf{u}|^2},\, 0.30\right). \tag{5}$$

Spin about $+\hat{\mathbf{z}}$ gives the special case $\hat{\mathbf{n}} \propto \hat{\mathbf{z}} \times \mathbf{u}$. Lift vanishes when the cross product is zero. Bray and Kerwin give a related vector model for free kicks [8].

2.3 Numerical Solution and Event Detection

The synthetic flight simulator uses fourth-order Runge–Kutta (RK4), a 2 ms step and a 1.5 s horizon. Linear interpolation locates $x = 11$ m and $z = R$, retaining goal-plane crossings before contact. Reintegrating 1,024 sampled launches per million-scale dataset at 1 ms leaves the crossing status unchanged and changes the recorded endpoints by at most about 0.012 mm. The ISSIA model-form check uses `solve_ivp`. Tracked-flight prefix forecasts use RK4 with a maximum 5 ms step.

2.4 Bayesian Inference of Launch Conditions

Let $\boldsymbol{\theta} = (\mathbf{v}_0, \boldsymbol{\omega}, C_D)$ and let $\mathbf{y}_1, \mathbf{y}_2$ be the two observed positions. With the start fixed and independent Gaussian errors,

$$p(\mathbf{y}_1, \mathbf{y}_2 \mid \boldsymbol{\theta}) = \prod_{i=1}^{2} \frac{1}{(2\pi\sigma^2)^{3/2}} \exp\left[-\frac{\|\mathbf{y}_i - \mathbf{r}(t_i; \boldsymbol{\theta})\|^2}{2\sigma^2}\right]. \tag{6}$$

Four-stage adaptive importance sampling combines a 20% prior component with a Gaussian proposal. The first three stages use possibly widened likelihoods to adapt the proposal by weighted moments. The fourth stage alone supplies the posterior at the specified σ, with weights proportional to likelihood times $p(\boldsymbol{\theta})/q(\boldsymbol{\theta})$. We use 40,000 draws per stage for the stated kick and report the conditional effective sample size (ESS). A weighted Gaussian kernel estimate on a 160×160 grid gives the mode and joint highest-density regions (HDRs) [24].

2.5 Million-Scale Simulation Dataset

We generate 1,225,000 launches per sampling strategy. Before flight integration, the full mixture assigns 40% to independent uniform draws, 24% to two DFL penalty anchors, 12% to three Asai curve-kick anchors and 24% to aerodynamic anchors from eleven other DFL segments, three ISSIA arcs and one digitised Barber–Carré trajectory. Each anchor stores fitted launch and aerodynamic parameters. For the five penalty and curve-kick anchors we perturb all seven fitted parameters, using a wider Gaussian for 30% of the draws. For the other fifteen anchors we draw launch speeds separately, transport the fitted spin into the sampled horizontal heading, and perturb the spin and drag. We clip to Table 1 and randomly mirror launches laterally, including the corresponding axial-spin components.

In the aimed version (v3), we select 65% of the non-Asai anchor draws with $v_x \geq 24\,\mathrm{m\,s^{-1}}$. For a nominal flight time $11/v_x$, we choose uniform ballistic targets in $Y \in [-3.85, 3.85]\,\mathrm{m}$ and $Z \in [0.55, 2.65]\,\mathrm{m}$ to set the transverse launch velocities. We rotate spin with the change in horizontal heading and integrate the full flight equations to obtain labels. The v2 comparison uses the same base mixture without ballistic aiming. For the two Asai-source holdouts, we replace all Asai-derived draws with the same independent uniform draws. The contest observations are used only for the final query and a descriptive count of nearby simulated prefixes.

Table 2: Airborne crossings by split, from 1,225,000 candidate launches per dataset. Splits are disjoint by trajectory and stratified by source anchor.

Dataset	Retained (%)	Train	Selection	Calibration	Test
v3, all anchors	89.07	872,841	54,553	54,550	109,108
v2, Asai held out	83.52	818,534	51,161	51,157	102,314
v3, Asai held out	87.44	856,879	53,556	53,553	107,112

The main v3 dataset retains 1,091,052 airborne crossings. The splits are assigned before perturbing the observations, with approximate proportions of 80%, 5%, 5% and 10%. At a six-coordinate Euclidean distance below 0.10 m from the contest prefix, the full mixture has 1,761 airborne neighbours. This count describes local sampling density. The unaimed source holdout falls below the design target of 85% airborne retention and is kept as the exploratory sampling control.

Our first models used SiLU on smaller simulations [23]. We then selected GELU in a model comparison and rebuilt the dataset and predictor using tracked-flight anchors, including Asai's measurements. The earlier datasets and model results are archived.

2.6 Wiener Residual Mixture Density Network

Let $\mathbf{b}_{\text{lin}}$ be a gravity-only crossing estimate based on midpoint velocity and let $\mathbf{b}_{\text{quad}}$ be a regularised quadratic estimate from the two observed positions and the known start. The feature extractor limits extrapolation times and uses the linear time estimate when the quadratic time is invalid. We blend the endpoints using a Wiener shrinkage weight,

$$\mathbf{b}_\sigma = \mathbf{b}_{\text{lin}} + w_\sigma(\mathbf{b}_{\text{quad}} - \mathbf{b}_{\text{lin}}), \qquad w_\sigma = \frac{s_*^2}{s_*^2 + 5\sigma^2/\Delta t^4}, \qquad s_* = 7.5\,\text{m}\,\text{s}^{-2}. \tag{7}$$

The denominator incorporates the second-difference acceleration variance (Appendix A). The fixed scale s_* sets the strength of shrinkage. The network uses 22 features, including velocities, acceleration, apparent drag and lift, extrapolation times and endpoints, w_σ and σ. The features are standardised using training data.

A 192-unit input projection and two 192-unit residual blocks use LayerNorm and GELU [21, 22]. The head produces five correlated bivariate Gaussian components,

$$p(\mathbf{q} \mid \mathbf{y}_1, \mathbf{y}_2, \sigma, E) = \sum_{k=1}^{5} \pi_k \mathcal{N}(\mathbf{q}; \mathbf{b}_\sigma + \boldsymbol{\delta}_k, \boldsymbol{\Sigma}_k), \qquad \mathbf{q} = (Y, Z). \tag{8}$$

The six outputs per component give a softmax weight logit, a two-dimensional residual $\boldsymbol{\delta}_k$, two positive scales $s_{kj} = \text{softplus}(r_{kj}) + 10^{-3}$ and correlation $\rho_k = 0.98\tanh(r_{k\rho})$. The diagonal of $\boldsymbol{\Sigma}_k$ is (s_{k1}^2, s_{k2}^2) and its off-diagonal entry is $\rho_k s_{k1} s_{k2}$.

Each dataset uses three runs (seeds 11, 42 and 155), AdamW with initial learning rate 0.002 and cosine decay, batch size 512 and 100 epochs [14]. Each epoch draws a new $\sigma \sim U[1, 18]$ mm per trajectory and adds independent Gaussian noise to the six coordinates. We choose the checkpoint by selection-set negative log-likelihood (NLL). On the independent calibration fold, separately at each test noise level, we fit a common scale multiplier by NLL and adjust the nominal 50% and 95% highest-density region (HDR) thresholds using ranks from 256 predictive draws per flight [24, 26]. We score physical-coordinate NLL, Euclidean errors of the predictive mean, and empirical coverage on the held-out test split. The data manifests record source and simulator hashes, split counts and dataset hashes. Computations use NumPy, SciPy, PyTorch and Matplotlib [27, 28, 29, 30].

2.7 Comparison Protocol

We repeat the model comparisons on fixed subsets of the v3 splits, drawing 65,536 training, 4,096 selection and 8,192 test trajectories without replacement (seed 20261108). Each candidate uses the same 60-epoch budget, three training seeds and optimizer, with the specified interventions below. Checkpoint selection minimises mean selection NLL at 2, 5, 10 and 15 mm. The full-mixture checkpoints supply the main forecast. They are scored on the same test subset as a reference trained with a larger budget.

Arms A and B use six coordinates plus σ. C and D use sixteen physical features, and E adds the quadratic endpoint to A's inputs. A and C predict absolute coordinates. B, D and E predict residuals about the quadratic endpoint, skip examples with no finite quadratic prediction during training, and use a zero endpoint in that case during evaluation. Arms A–E use the same three-layer, width-192 GELU body and five components. We also retrain the legacy coordinate MDN (two 64-unit SiLU layers, three components) and PI-ResMDN (three 128-unit SiLU layers, five components and its original 16 features and quadratic fallback) on these v3 rows.

For architecture screening, we compare 11 and 16 physical inputs for the direct target, and vary mixture size and depth using the 22-feature Wiener residual target. Depth is screened with

SiLU. A shallower candidate is preferred when its NLL disadvantage is within one paired seed standard deviation and the mean absolute HDR coverage gap rises by at most 1.5 percentage points. We then compare activations at that depth. The intervention screen holds the three-layer v3 architecture fixed and changes only covariance correlation, the learning rate schedule, or the reuse of the initial noise draw for each trajectory.

3 Results

3.1 Answer for the Stated Kick

The quadratic baseline gives $T = 0.405\,\text{s}$ and $(Y, Z) = (2.495, 1.920)\,\text{m}$ (Appendix A). At $\sigma = 5\,\text{mm}$, the calibrated full-mixture MDN (seed 11) gives a mode of $(2.666, 1.792)\,\text{m}$ and a mean of $(2.711, 1.859)\,\text{m}$. The mode heights for seeds 42 and 155 are approximately 1.83 and 1.80 m. The uniform-prior Bayesian reference gives a mode of $(2.650, 2.052)\,\text{m}$ and joint region areas of 0.219 and $0.809\,\text{m}^2$ (ESS 2,893). The two models share flight equations but use different launch distributions and density estimators. Their modes differ in height by about 0.26 m.

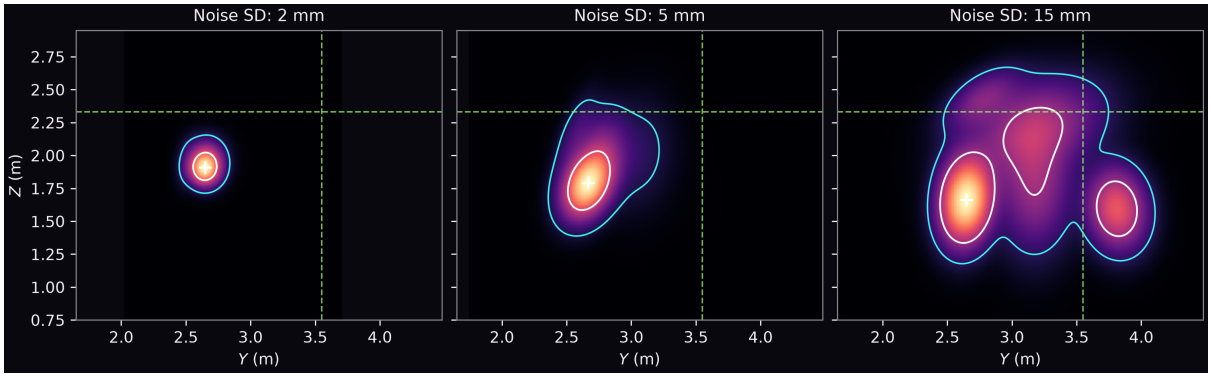


Figure 1: Seed-11 density near the right side of the goal. White and cyan enclose the calibrated 50% and 95% regions. The dashed green lines mark the radius-inset boundaries and the white cross marks the global density mode.

Table 3: Calibrated seed-11 MDN for the stated kick, conditional on airborne crossing. Areas come from numerical integration on a two-dimensional density grid. In-frame probabilities integrate the Gaussian mixture over the inset rectangle.

Noise (mm)	Mode Y (m)	Mode Z (m)	50% area (m^2)	95% area (m^2)	In-frame
2	2.648	1.910	0.030	0.137	$> 99.9\%$
5	2.666	1.792	0.106	0.619	97.7%
10	2.656	1.684	0.334	1.585	84.1%
15	2.647	1.664	0.532	1.940	69.7%

At 5 mm, the seed-11 MDN's 5th-to-95th percentile widths are 0.555 m in Y and 0.682 m in Z, compared with 0.601 and 0.830 m for the physics posterior. Using the same inset rectangle, their in-frame probabilities are 97.7% and 91.1%. The respective masses above the inset crossbar height are 2.31% and 8.90%. Lateral misses account for less than 0.03% in either model.

The physics sample shows how similar prefixes can lead to different crossings (Fig. 2). Draws A and B fit the six coordinates within 4 mm RMSE and cross at nearly the same Y, but differ by 0.29 m in height and have opposite vertical Magnus accelerations. Draw C has weaker leftward

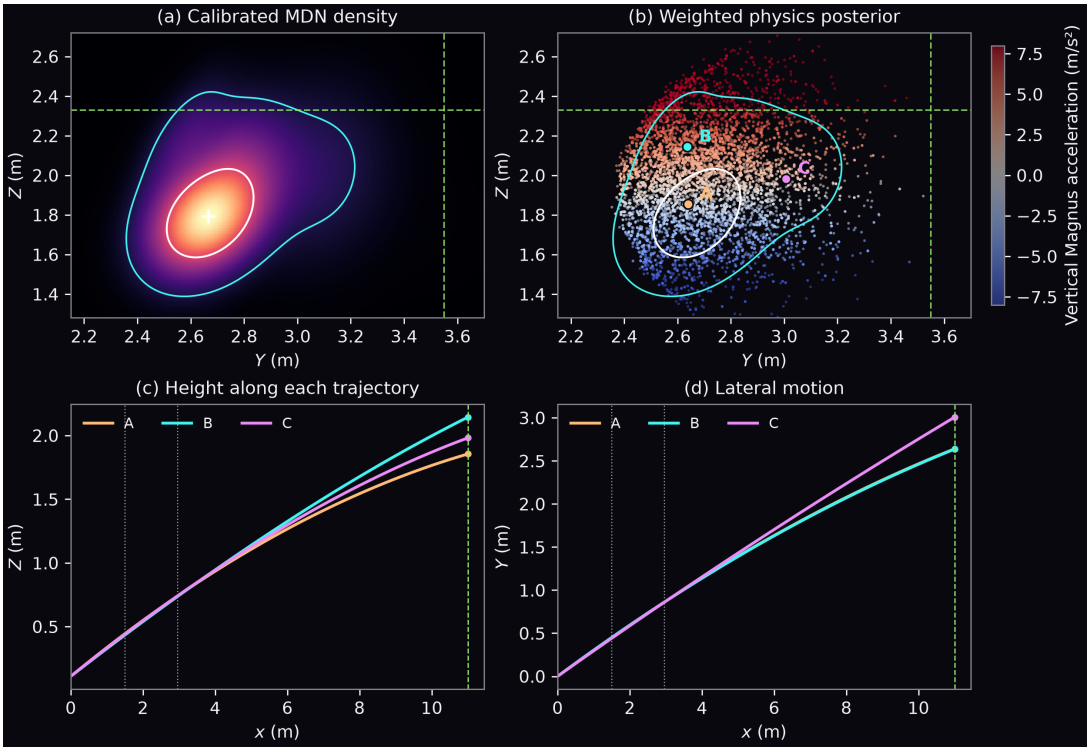


Figure 2: Physical interpretation at 5 mm. (a) Calibrated v3 density. (b) Weighted resamples from the physics posterior, coloured by vertical Magnus acceleration at the observed midpoint velocity, with the same v3 contours overlaid. (c, d) Representative trajectories. The dotted lines mark the observed longitudinal positions and the dashed green line marks the plane.

lift and crosses farther right. The component responsibilities in Table 4 describe how the seed-11 mixture represents these endpoints.

Table 4: Representative physics posterior draws. Magnus acceleration is evaluated along each trajectory at 0.05 s in $\mathrm{m\,s^{-2}}$. Responsibilities are in mixture component order.

Draw	RMSE (mm)	(Y, Z) (m)	$a_{\mathrm{M},y}$	$a_{\mathrm{M},z}$	Responsibilities
A	3.25	(2.64, 1.86)	−9.62	−1.22	(0.001, 0.108, 0.000, 0.891, 0.001)
B	3.93	(2.64, 2.14)	−10.57	+5.34	(0.004, 0.531, 0.000, 0.209, 0.256)
C	4.97	(3.01, 1.98)	−3.84	+1.48	(0.562, 0.347, 0.001, 0.090, 0.000)

Multimodality. At 5 mm, the Z marginal and the vertical slices at $Y = 2.62$ and 2.67 m are unimodal in all three v3 fits. Seed 155 nevertheless has a second joint-density maximum near (2.55, 2.22) m. A slice through one lateral coordinate can miss a separate joint peak. At 15 mm, each fit has two slice peaks near $Z = 1.64$–1.72 and 2.33–2.38 m, while every Z marginal still has one maximum (Fig. 3). The physics posterior's slice at $Y = 2.67$ m has one peak at Scott's KDE bandwidth and two at 0.75 times that bandwidth. The number of peaks depends on the assumed noise, density representation and choice of slice (Appendix B).

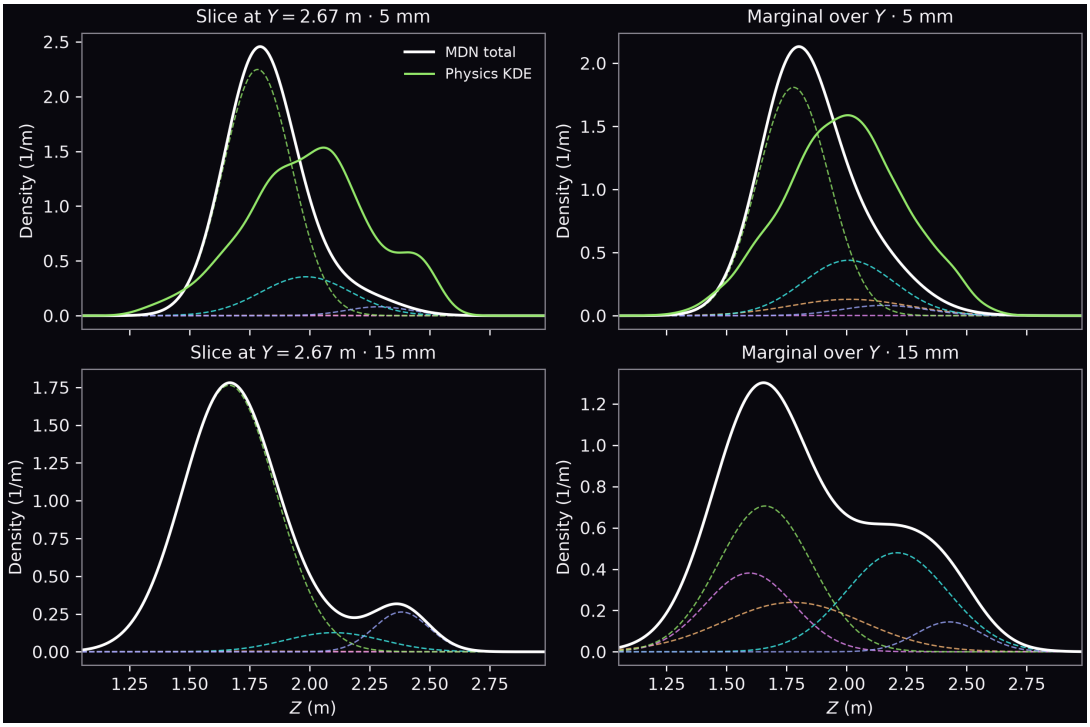


Figure 3: Seed-11 vertical slices and marginals. The white curve is the total MDN density and dashed curves show its five weighted Gaussian components. Green shows the weighted physics KDE at 5 mm with Scott's bandwidth. The fixed-Y slices are normalised to integrate to one over height.

3.2 Feature, Target and Architecture Comparisons

Physical features improve the direct model at 5 mm (C versus A in Table 5). The matched-budget Wiener model has the lowest mean test NLL among the retrained models at all four noise levels. Its median error at 5 mm is 0.269 m, close to PI-ResMDN's 0.270 m. Arm D reaches NLL 2.639 at 15 mm. The quadratic predictor has no finite forward root on 466 of the 8,192 test trajectories at this noise level, so D uses its zero-endpoint fallback on those rows.

Table 5: Model comparison at 5 mm on the same 8,192 v3 test trajectories and perturbations. Results average three seeds. Only the last row uses full training and calibration.

Model	Median (m)	P90 (m)	NLL
Legacy coordinate MDN	0.297	0.623	−0.073
A: coordinates, direct	0.282	0.603	−0.116
B: coordinates, quadratic residual	0.275	0.604	−0.140
C: physics features, direct	0.274	0.599	−0.173
D: physics features, quadratic residual	0.295	0.633	0.208
E: coordinates + quadratic endpoint, residual	0.274	0.607	−0.146
PI-ResMDN, retrained	0.270	0.598	−0.269
Wiener-ResMDN, matched budget	0.269	0.599	−0.300
Wiener-ResMDN, full training and calibration	0.266	0.597	−0.376

In the two-layer SiLU screen, the 11- and 16-feature direct variants give selection NLLs of −0.145 and −0.159. Holding the Wiener input and residual target fixed, one, three and five mixture components give −0.014, −0.186 and −0.238. Four layers win the SiLU depth screen. At depth four, GELU gives the lowest mean NLL (−0.253), followed by Tanh (−0.249), SiLU (−0.247)

and ReLU (−0.222), with variation across seeds (Fig. 4). This screen uses the smaller, matched training budget.

In the three-layer GELU intervention screen, the reference mean selection NLL is −0.252. Removing within-component correlation, fixing the learning rate, and reusing one set of training perturbations per trajectory raise it to −0.225, −0.208 and −0.168, respectively.

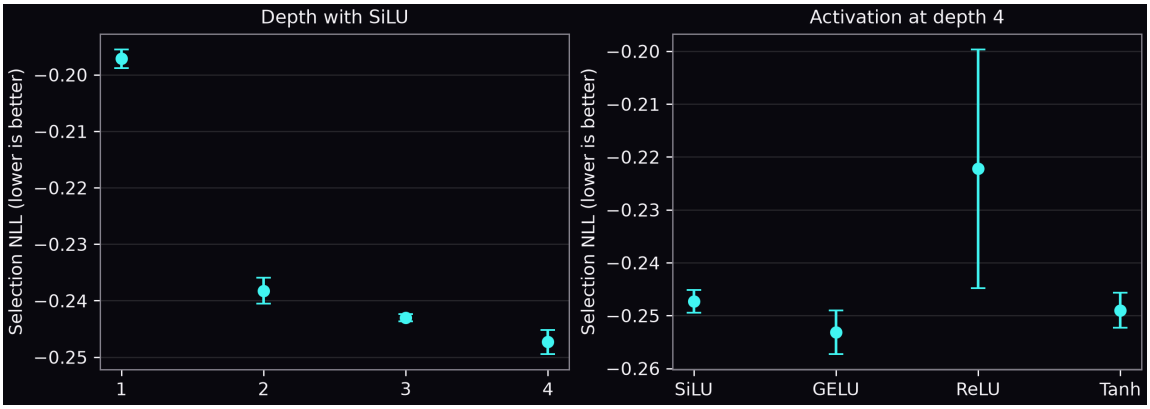


Figure 4: Depth and activation screens with the v3 Wiener residual inputs. Each point is the mean selection NLL across four noise levels and three training seeds. Error bars are sample standard deviations across seed means.

3.3 Holdout Accuracy and Calibration

Table 6 scores the 109,108 held-out full-mixture crossings. At 5 mm the MDN median error is 0.266 m and the quadratic baseline's is 0.431 m. The baseline uses identical coordinate perturbations and its error is computed on the subset with a finite forward root. The counts without a finite forward root are 0, 102, 2,139 and 6,003 at the four noise levels. The MDN produces a density for every test prefix.

Table 6: Full-mixture v3 test results. Calibrated MDN scores are means over three training seeds. Point errors are in metres. HDR coverage is in percent.

Noise (mm)	Quad. median	MDN median	MDN P90	MDN NLL	50% HDR	95% HDR
2	0.188	0.126	0.292	−1.766	50.16	95.27
5	0.431	0.266	0.597	−0.371	50.08	95.08
10	0.843	0.416	0.802	0.332	50.58	95.32
15	1.235	0.488	0.868	0.590	50.44	95.42

Table 7 and Fig. 5 separate scale fitting from HDR rank adjustment. Scale fitting changes coverage by at most 0.14 percentage points. At 15 mm, the full calibration raises inner coverage from 49.33% to 50.44% and lowers outer coverage from 95.87% to 95.42%. The corresponding outer median area, averaged across seeds, drops from 2.018 to 1.975 m^2.

Conditional in-frame expected calibration errors are 0.0032, 0.0039, 0.0044 and 0.0056 in increasing noise order. The Brier scores are 0.0226, 0.0451, 0.0644 and 0.0731. The observed in-frame fraction among simulated airborne crossings is 56.5%.

Table 7: Calibration stages on the full test split. Coverage entries give the 50% / 95% regions in percent. The last column gives calibrated median areas in m^2. All entries are averaged across seeds.

Noise (mm)	Raw coverage	Scale only	Scale + HDR	Final areas
2	50.15 / 95.22	50.29 / 95.27	50.16 / 95.27	0.040 / 0.201
5	49.77 / 95.44	49.81 / 95.45	50.08 / 95.08	0.171 / 0.854
10	49.71 / 95.74	49.65 / 95.72	50.58 / 95.32	0.375 / 1.659
15	49.33 / 95.87	49.26 / 95.84	50.44 / 95.42	0.497 / 1.975

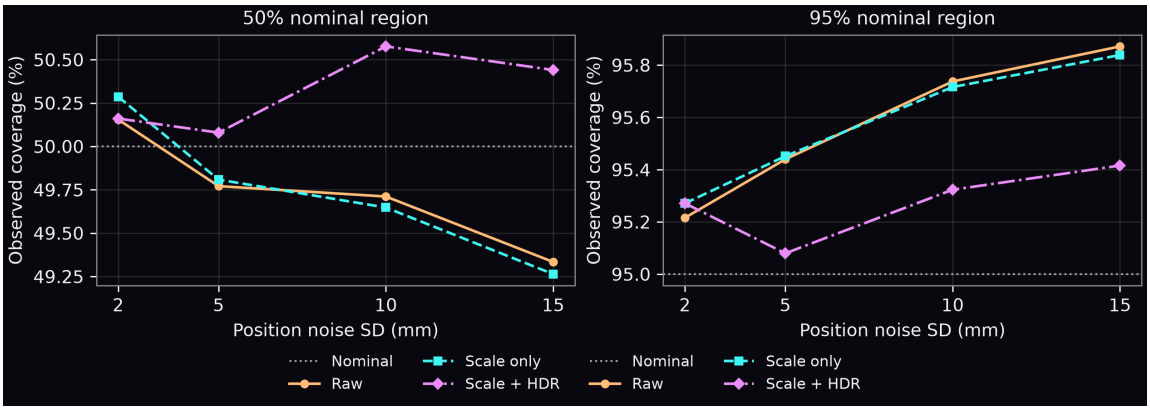


Figure 5: Observed joint-region coverage before calibration, after scale fitting, and after HDR rank adjustment on 109,108 held-out v3 trajectories. The dotted lines give the nominal levels.

3.4 Paired Comparison with the Physics Posterior

We draw 32 trajectories from the v3 test split (seed 155108) and use the same 5 mm perturbation for each method. Adaptive importance sampling starts with 6,000 proposals per stage and retries with 18,000 when conditional ESS falls below 100. After retry, 30 trajectories meet the ESS threshold and have a finite quadratic prediction. The median Euclidean errors are 0.478 m for quadratic extrapolation, 0.314 m for the Bayesian reference and 0.270 m for the calibrated v3 model (seed 11).

Median energy scores are 0.241 and 0.173 for the Bayesian reference and v3 model, respectively. The energy score is $\mathbb{E}\|\mathbf{Q}-\mathbf{q}\| - \frac{1}{2}\mathbb{E}\|\mathbf{Q}-\mathbf{Q}'\|$, with independent predictive draws $\mathbf{Q}, \mathbf{Q}'$ and truth $\mathbf{q}$ [25]. Median CPU times are 2.19 s and 1.74 ms, including inference, sampling or resampling 4,096 endpoints, and scoring. The small ESS-filtered subset and the different prior distributions limit the accuracy comparison.

3.5 Aimed Sampling and Asai Source Holdout

With Asai withheld, aiming increases the count of airborne endpoints with $v_x \geq 24\,\mathrm{m\,s^{-1}}$ within the sampling grid from 483,698 to 550,313 (13.8%). The grid spans $|Y| \leq 3.66$ m and $R \leq Z \leq 2.44$ m in 6×3 cells. Its least populated cell grows from 20,207 to 24,984. This design diagnostic uses the full lateral width and crossbar height.

On 53,126 identical uniform-prior test trajectories with matched 5 mm perturbations, median error changes from 30.26 to 30.09 cm, while NLL changes from -0.2602 to -0.2577. Aiming changes both the distribution and the number of usable training trajectories. The comparison estimates their combined effect.

The Asai free kicks were tracked at 100 Hz with the goal at 23 m [5]. We take positions at 0.05 and 0.10 s relative to the initial frame, shift the starting height to R for inference, then restore the height offset and score against an interpolated crossing at a virtual 11 m plane. Table 8 uses the two models whose priors exclude these trials. It uses an assumed $\sigma = 10$ mm and the original measured positions. The full-mixture model uses the same kicks as prior anchors, so the source-holdout results apply to the two reduced mixtures.

Table 8: Asai source holdout. Euclidean errors of the predictive mean, in centimetres, averaged over three training seeds.

Sampling strategy	Fig3	Sample05	Sample12	Mean over kicks
v2, unaimed	56.53	56.73	61.73	58.33
v3, aimed	57.59	57.51	62.28	59.13

The three observed kicks show no transfer gain from aiming under this noise assumption. The trial count and use of a virtual crossing plane limit the scope of the comparison.

3.6 Distribution Shift

We take 6,000 v3 test launches (seed 155107), retain paired airborne crossings under each perturbation, and reuse the same 5 mm noise. With the seed-11 model and calibrator fixed, a $3\,\mathrm{m\,s^{-1}}$ crosswind (5,978 pairs) lowers 50% coverage from 50.7% to 42.9% and 95% coverage from 95.0% to 91.6%. Spin decay with a 0.5 s time constant (5,971 pairs) lowers inner coverage from 50.2% to 39.3% while median error falls from 0.264 to 0.233 m. Smaller point error can coincide with worse coverage.

3.7 Rebound and Rolling Crossings

For low shots, we continue the flight when the ball centre reaches $z = R$. The vertical velocity reverses with normal restitution $e_n = 0.60$. At impact, the horizontal slip is $\mathbf{v}_t + (\boldsymbol{\omega} \times (-R\hat{\mathbf{z}}))_t$. With thin-shell inertia $I = 2mR^2/3$, the tangential impulse opposes slip and is capped at μ_c times the normal impulse ($\mu_c = 0.40$). The impulse changes both horizontal velocity and spin. When the incoming normal speed drops below $0.50\,\mathrm{m\,s^{-1}}$, the ball rolls with resistance $\mu_r g$ ($\mu_r = 0.08$) and quadratic drag. We extend the integration horizon to 3 s. These coefficients define the simulated surface.

We use a separate 100,000-launch v3 bank with the full anchor mixture. The simulator preserves the 88,999 direct crossings and recovers 11,001 additional crossings (6,636 during a rebound and 4,365 while rolling). Of the post-contact crossings, 7,761 are in-frame under the radius-inset criterion. The contact simulator supplies positions after impact if an observation occurs after the first contact. Varying e_n over 0.56–0.67, μ_c over 0.25–0.55 and μ_r over 0.046–0.153 one at a time leaves 11,001 post-contact crossings in each simulated scenario. The in-frame count ranges from 7,742 to 7,761.

Two Wiener residual MDNs have the same architecture as Section 2.6, seed 11 and 100 training epochs. The airborne-only model trains on 71,202 direct crossings. The contact-inclusive model trains on the same rows plus 8,798 post-contact crossings. Each model uses its own selection and calibration folds from the same partition. Both are evaluated on the same 10,000 rows with identical noise and receive only the two positions and σ through the feature extractor. Ground-contact status defines evaluation subsets. The contact-inclusive model estimates the endpoint density conditional on crossing by either route.

Table 9: Paired rebound experiment at $\sigma = 5\,\text{mm}$. In each cell, the first number is the airborne-only model and the second is the contact-inclusive model. Coverage is in percent.

Test subset	Median (m)	NLL	50% HDR	95% HDR
Direct ($n = 8{,}899$)	0.271 / 0.272	−0.288 / −0.268	49.7 / 49.0	95.1 / 94.7
Post-contact ($n = 1{,}101$)	0.242 / 0.163	+0.053 / −3.181	30.8 / 58.9	84.1 / 97.5
All ($n = 10{,}000$)	0.267 / 0.261	−0.251 / −0.588	47.6 / 50.2	94.0 / 95.3

At 5 mm, a paired row bootstrap gives a post-contact median error difference of $-0.080\,\text{m}$ (95% interval $[-0.089, -0.069]\,\text{m}$) for contact-inclusive minus airborne-only training. The direct-crossing difference is $+0.001\,\text{m}$ ($[-0.001, +0.003]\,\text{m}$). The post-contact 50% region still overcovers at 58.9%, and 22.0% of its predictive samples lie below $Z = R$ because the Gaussian mixture is untruncated. A 200-epoch airborne-only control gives a post-contact median error of 0.244 m and outer coverage of 80.3% at the same noise.

For the stated kick, the two models give modes of $(2.661, 1.798)$ and $(2.655, 1.793)\,\text{m}$ at 5 mm. A separate importance-sampling audit with the uniform reference prior assigns numerically negligible mass to first contact before crossing. At 5 mm the two runs have ESS 2,132 and 1,767. These finite samples support treating the case as a direct crossing in this model. They provide no quantitative upper bound on the rare-event probability.

3.8 Sensitivity to the Drag Prior

For the stated kick, raising the upper bound of C_D from 0.30 to 0.45 shifts the posterior mean by about 2 cm. Mean 50% and 95% areas change by +1.1% and −0.5%, respectively, less than the variation across the three seeds.

Table 10: Sensitivity of the case posterior to the upper limit of the drag prior ($\sigma = 5\,\text{mm}$, three seeds, 40,000 samples per stage). Ranges are minima and maxima over seeds; means are averages over seeds.

C_D prior	50% area (m^2)	95% area (m^2)	Mean (Y, Z) (m)
$U[0.15, 0.30]$	0.213 (0.205–0.219)	0.806 (0.789–0.821)	(2.696, 1.988)
$U[0.15, 0.45]$	0.215 (0.210–0.222)	0.802 (0.792–0.809)	(2.689, 1.971)

3.9 Model-Form Check on Tracked Arcs

We fit the first six frames of ISSIA-3D segments 608–622 and 1144–1172, sampled at 25 fps, and score the remaining frames [15, 16]. Selection required at least two cameras, eight consecutive frames, height of at least 0.30 m and median reprojection error below 10 pixels per frame. Absolute 3D accuracy and action labels were unverified for these segments.

Drag reduces held-out RMSE in both arcs. Adding an effective lift component in the vertical plane gives the lowest errors (Table 11). The fitted drag and lift have large uncertainties and may absorb tracking or model error. These suffix forecasts check the flight equations. Related ISSIA arcs and the DFL penalties below also contribute to the main simulation prior.

Table 11: Held-out RMSE and last-frame error, in metres, for two ISSIA arcs. Initial conditions and any free aerodynamic coefficients use the first six frames.

Arc	Quantity	Gravity	$C_D = 0.22$	$C_D = 0.45$	Drag + lift
608–622	Forecast RMSE	0.697	0.489	0.469	**0.188**
	Final error	1.243	0.863	0.811	**0.308**
1144–1172	Forecast RMSE	2.179	1.546	1.584	**1.084**
	Final error	4.169	2.876	2.942	**2.119**

3.10 Prefix Forecasts on Tracked Flights

We fit the first three DFL positions at 0, 0.04 and 0.08 s and forecast the held-out trajectory in local coordinates [17]. Both kicks resulted in goals. The start frames are 128955 and 176244 (events 18232100000995 and 18242100001386). With 15 mm assumed noise, unknown spin improves one crossing estimate and slightly worsens the other (Table 12).

Unknown spin raises held-out RMSE from 0.338 to 0.366 m on ISSIA frames 2972–2991. For 22 held-out planar positions digitised from Barber and Carré's Fig. 5, RMSE rises from 2.266 to 2.301 m at an assumed 67 mm digitisation scale [18].

Table 12: Two DFL penalties with seven scored positions before each reference crossing. Both trajectory RMSE and crossing-point error are in metres. The crossing forecasts condition on airborne arrival at each observed goal plane.

	Trajectory RMSE		Crossing error	
Case	Zero spin	Unknown spin	Zero spin	Unknown spin
J03WOH	0.347	0.292	0.645	0.543
J03WR9	0.309	0.305	0.542	0.553

Of 60 DFL flight candidates, 11 passed motion and event screening. At both 15 and 30 mm, unknown spin lowers held-out RMSE in 7 of 11 segments, with median paired improvements of 0.027 and 0.011 m. The screen fits a quadratic to the full segment, including later frames, and favours smooth arcs. These segments lack goal-plane labels.

4 Discussion

Two noisy positions constrain average velocity more tightly than acceleration. The Wiener anchor shrinks the quadratic estimate more strongly as σ increases. The mixture then learns how to correct that anchor under the training distribution. The wind and spin-decay tests show that nominal HDR coverage depends on how closely the flight dynamics match that distribution.

The aimed prior gives better representation of fast in-frame trajectories. Its small shared-test gain and the three Asai replays put the coverage gain in perspective. The replay results depend on the assumed noise and the fixed observation schedule. The DFL and ISSIA checks address individual physical trajectories, and calibration for real penalties remains unmeasured.

The airborne model forecasts geometric in-frame arrival conditional on E. The contact-inclusive model conditions on crossing by either route within the simulated horizon. An unconditional in-frame forecast requires a separate estimate of crossing probability. The geometry accounts for ball radius and omits frame contact and goalkeeper intervention. The post-contact comparison depends on the assumed surface parameters and simulated mix of rebounds and rolls.

In a local calculation on held-out v3 launches, an additional 0.15 s observation with the same 5 mm noise reduces the median radial crossing spread from 0.284 to 0.113 m under a Gaussian reference prior (Appendix A.3). This measurement design adds 50 ms before the forecast can be made.

5 Conclusion and Outlook

At an assumed 5 mm coordinate noise, the million-trajectory MDN predicts a most likely crossing at $(2.67, 1.79)$ m for the contest kick. Its 95% region covers 95.1% of the held-out still-air simulations at this noise level. Crosswind reduces coverage under the fixed model. Including post-contact trajectories substantially improves simulated rebound and rolling forecasts, with a 1 mm median difference on direct crossings in the paired experiment. High-rate tracking with independently measured noise and a real holdout spanning direct crossings, misses and ground contact would test these findings.

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A Constant-Acceleration Baseline and Noise Propagation

A.1 Baseline from the Three Given Positions

For the positions in Eq. (1) and $\Delta t = 0.05\,\mathrm{s}$, the midpoint velocity and acceleration are

$$\mathbf{v}(\Delta t) = \frac{\mathbf{r}_2 - \mathbf{r}_0}{2\Delta t} = (29.6, 8.6, 6.3)\,\mathrm{m\,s^{-1}}, \tag{9}$$

$$\mathbf{a} = \frac{\mathbf{r}_2 - 2\mathbf{r}_1 + \mathbf{r}_0}{\Delta t^2} = (-16, -16, -12)\,\mathrm{m\,s^{-2}}. \tag{10}$$

Writing $\tau = T - \Delta t$, solve $-8\tau^2 + 29.6\tau - 9.5 = 0$ and take the earlier forward root, $\tau = 0.355\,\mathrm{s}$. Substitution gives $T = 0.405\,\mathrm{s}$ and

$$(Y, Z) = (0.45 + 8.6\tau - 8\tau^2,\ 0.44 + 6.3\tau - 6\tau^2) = (2.495, 1.920)\,\mathrm{m}. \tag{11}$$

The benchmark baseline selects the earliest finite quadratic root after 0.10 s, with a linear branch when a_x is near zero.

A.2 Propagation of Position Noise to the Endpoint

With an exact start and two independent coordinate errors of variance σ^2, the second difference has error $\boldsymbol{\varepsilon}_2 - 2\boldsymbol{\varepsilon}_1$ and component variance $5\sigma^2$ [20]. Thus

$$\sigma_a = \frac{\sqrt{5}\,\sigma}{\Delta t^2} = 4.47\,\mathrm{m\,s^{-2}} \quad \text{at } \sigma = 5\,\mathrm{mm}. \tag{12}$$

An independently noisy start would raise the component variance to $6\sigma^2$.

For a fixed crossing time T, let $n = T/\Delta t$. In the quadratic interpolant through the exact start and the two observations, the noisy positions have Lagrange weights $L_1 = -n(n-2)$ and $L_2 = n(n-1)/2$. The per-axis standard deviation at fixed T is $\Lambda(n)\sigma$, where

$$\Lambda(n) = \sqrt{L_1^2 + L_2^2}, \qquad e_{50}^{\mathrm{fixed}} = \sqrt{2\ln 2}\,\Lambda(n)\sigma. \tag{13}$$

Here e_{50}^{fixed} is the median radial deviation in (Y, Z) for isotropic Gaussian errors. The stated kick has $n = 8.10$ and $\Lambda = 57.2$, giving 0.337 m at 5 mm.

We also propagate the observations through 150,000 quadratic fits per noise level, solving for the crossing time in each draw. This diagnostic requires a positive midpoint-relative root, $T \le 1.5\,\mathrm{s}$ and $Z > R$ at the plane. At 5 mm its median radial deviation from the unperturbed quadratic crossing is 0.339 m.

The spreads in Table 13 describe perturbations of the stated case. Table 6 reports Euclidean prediction errors across simulated flights.

Table 13: Constant-acceleration propagation of observation errors. Intervals contain the 5th to 95th percentiles of retained draws.

σ (mm)	Discarded	s.d. T (ms)	s.d. Y (m)	s.d. Z (m)	Y 5–95% (m)	Z 5–95% (m)
2	0	4.8	0.115	0.115	2.31–2.68	1.73–2.11
5	0	12.3	0.290	0.289	2.02–2.97	1.45–2.40
15	5,955	41.6	0.933	0.871	1.02–4.04	0.62–3.48

A.3 Measurement Design and Local Crossing Spread

We sample 4,000 airborne launches without replacement from the full-mixture v3 test split (seed 155107). Let $\mathbf{h}(\boldsymbol{\theta})$ be the observed positions and $\mathbf{q}(\boldsymbol{\theta}) = (Y, Z)$ the crossing. We compute their Jacobians J_h and J_q by centred differences about each simulated launch. With a diagonal Gaussian reference covariance Σ_θ whose variances match the uniform ranges, the local crossing covariance is

$$\Sigma_q = J_q(\sigma^{-2} J_h^\top J_h + \Sigma_\theta^{-1})^{-1} J_q^\top, \tag{14}$$

Each prior variance is $(\theta_{\max} - \theta_{\min})^2/12$. We estimate each flight's median radial spread from 64 draws of $\mathcal{N}(\mathbf{0}, \Sigma_q)$ and report the median across flights in Table 14. The third-position variant adds a reading at 0.15 s with independent noise of the same standard deviation. These are local uncertainty estimates under the simulated dynamics and the Gaussian reference prior, which differs from the mixture used for training.

Table 14: Local median radial crossing spread (m) over 4,000 v3 test launches.

σ (mm)	0.05, 0.10 s	0.05, 0.10, 0.15 s
2	0.131	0.049
5	0.284	0.113
10	0.432	0.207
15	0.503	0.282

B Case Decomposition Details

We refine joint maxima from a 1201 × 1201 grid, retaining those above 0.2% of the global peak. Slice and marginal peaks require a prominence of at least 0.2% of their profile maximum. Reported peaks are at least 0.05 m apart.

Table 15: Joint maxima counts and vertical slice and marginal peak heights (m).

Noise (mm)	Seed	Joint maxima	Slice $Y = 2.62$	Slice $Y = 2.67$	Marginal Z
5	11	1	1.77	1.79	1.80
5	42	1	1.83	1.83	1.84
5	155	2	1.79	1.80	1.81
15	11	4	1.66, 2.34	1.67, 2.36	1.65
15	42	4	1.69, 2.33	1.72, 2.35	1.69
15	155	4	1.64, 2.35	1.65, 2.37	1.64

The physics posterior retains 22,583 airborne draws (seed 8047, ESS 2,893). Its weighted correlations are $\mathrm{corr}(\omega_y, Z) = -0.911$, $\mathrm{corr}(\omega_z, Y) = +0.908$ and $\mathrm{corr}(C_D, Y) = -0.046$. Each responsibility is the weighted component density divided by the mixture density at the given endpoint.